Regularized Multivariate Two-way Functional Principal Component Analysis

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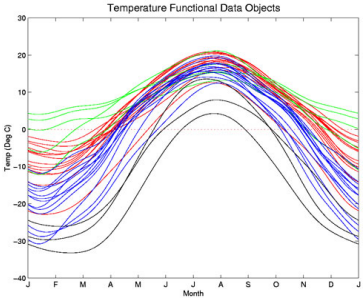
Outline

1. Introduction & Background
2. A SVD Approach for Regularized Multivariate FPCA
3. Two-way Regularized Multivariate FPCA
4. Conclusion & Future Work

Introduction

- **Functional data** arise when measurements vary continuously over a domain such as time or space, and are naturally viewed as curves rather than isolated points.



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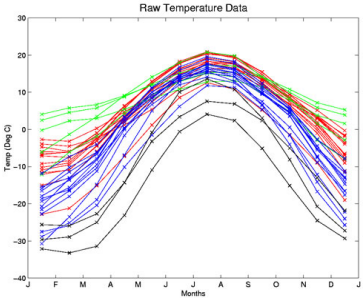

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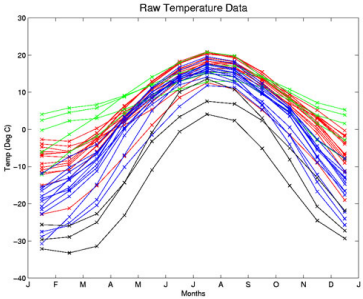

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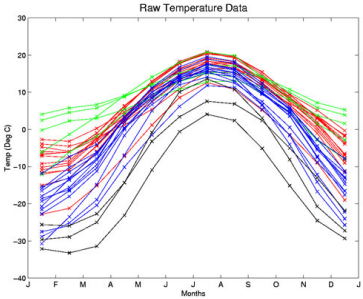
- **Functional Data Analysis (FDA)** models these observations as smooth underlying functions, improving interpretation and analysis of continuous variation.
- **Functional PCA (FPCA)** extends classical PCA to functional data by identifying orthogonal functional components that capture the main modes of variation for dimension reduction [@ramsay2005functional].



Functional Principal Component Analysis

- **Extensions of FPCA:**
  - Smoothed FPCA: Adds roughness penalties [Silverman, 1996, Huang et al., 2008].
  - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].
  - Multivariate FPCA (MFPCA) [Silverman, 1996, Happ and Greven, 2018].
  - Regularized MFPCA: Penalties improve estimation & interpretability [Haghbin et al., 2025].
  - **Two-way functional data:** consist of a data matrix whose row and column domains are both structured. Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- **Impact:** More adaptable, robust, and applicable across different problems.

Foundations of FPCA through Minimizing Reconstruction Error

- **Goal:** Capture the dominant functional variation in $X \in \mathbb{R}^{n \times m}$ using a low-rank representation, where n is the number of discretized functions and m is the number of grid points. The vector $v \in \mathbb{R}^m$ represents the principal component function, while $u \in \mathbb{R}^n$ gives the corresponding scores.
- **Reconstruction problem:**
$$\min_{u,v} \|X - uv^\top\|_F^2 = \min_{u,v} \operatorname{tr}\{(X - uv^\top)(X - uv^\top)^\top\}.$$
- **Extensions of FPCA via Regularization:**
$$\min_{u,v} \|X - uv^\top\|_F^2 + \mathcal{P}(u, v)$$
  - **Smooth FPCA:** adds roughness penalty on functions.
  - **Sparse FPCA:** adds sparsity penalty on loadings.
- **Algorithms:** Based on iterative **power method** and **thresholding** updates.

Smooth Functional PCA

- **Problem setup** [Huang et al., 2008]: Smooth FPCA estimates a low-rank representation of $X \in \mathbb{R}^{n \times m}$ by solving

$$\min_{u,v} \|X - uv^\top\|_F^2 + \alpha\, v^\top \Omega v,$$

where $u \in \mathbb{R}^n$ denotes the scores, $v \in \mathbb{R}^m$ the loading function, Ω the roughness penalty matrix, and α the smoothing parameter. Define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$.

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Algorithm: Penalized Power Iteration

1. Initialize v.
2. Repeat until convergence:
   1. $u \leftarrow Xv$
   2. $v \leftarrow S(\alpha) X^\top u$
   3. $v \leftarrow \dfrac{v}{\|v\|}$
3. Update $X \leftarrow X - \sigma\, uv^\top$ and proceed to next component.

Sparse Functional PCA

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Algorithm: sFPCA-rSVD

1. Initialization: Compute the best rank-one approximation of X using SVD
2. Iterate until convergence:
   1. Update Left Singular Vector: $u \leftarrow Xv$
   2. Update Right Singular Vector: $v \leftarrow h_\gamma X^\top u$
   3. Normalize Right Singular Vector: $v \leftarrow \frac{v}{\|v\|}$

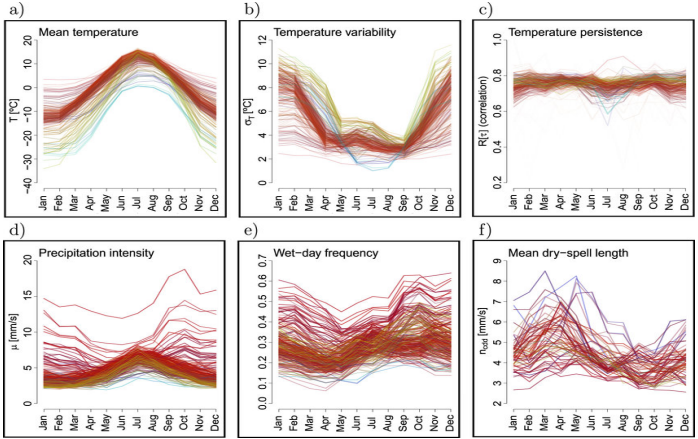
Overview of Existing Approaches

- **Smooth FPCA:**
  - Pros: Produces smooth eigenfunctions.
  - Algorithm: Penalized power iteration.
  - Tuning: α (smoothness) via GCV.
- **Sparse FPCA:**
  - Pros: Feature selection $\rightarrow$ interpretable.
  - Algorithm: sFPCA-rSVD algorithm.
  - Tuning: γ (sparsity) via CV.
- **Combined Approaches:** Smooth + Sparse together.

$$\min_{u,v} \|X - uv^\top\|_F^2 + \alpha v^\top \Omega v + p_\gamma(v)$$

Regularized MFPCA

- **Context**
  - Univariate FDA → **Multivariate FDA** → joint modes of variation across functions



Regularized MFPCA

- **Context**
  - Univariate FDA $\rightarrow$ **Multivariate FDA** $\rightarrow$ joint modes of variation across functions
- **Challenges**
  - Discretization & irregular grids $\rightarrow$ noise, missing data
  - High dimensionality and limited sample size $\rightarrow$ unstable eigenfunctions
  - Cross-function correlation $\rightarrow$ requires enforcing smoothness both within and across functions
- **Proposed Solution: Penalized SVD**
  - **Smoothness penalties**: roughness on derivatives
  - **Sparsity penalties**: Soft, hard, or SCAD
  - **Block-diagonal roughness matrix** for cross-function structure
- **Impact**
  - Produces **smooth, sparse, interpretable** joint modes
  - More stable & applicable to high-dimensional multivariate FDA

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating** p **functional data matrices**.
  - Each variable: $X_j \in \mathbb{R}^{n \times m_j}$ where n: number of observations and m_j: grid points
- **Rank-one approximation** (per variable):

$$X_j \approx u_j v_j^\top, \quad u_j \in \mathbb{R}^n, \; v_j \in \mathbb{R}^{m_j}$$

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- **Full data matrix**: $\boldsymbol{X} = \begin{bmatrix} X_1 & X_2 & \cdots & X_p \end{bmatrix}$

$$\boldsymbol{X} = \begin{bmatrix} x_{11}(t_{11}) & \cdots & x_{11}(t_{1m_1}) & \cdots & x_{1p}(t_{p1}) & \cdots & x_{1p}(t_{pm_p}) \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1}(t_{11}) & \cdots & x_{n1}(t_{1m_1}) & \cdots & x_{np}(t_{p1}) & \cdots & x_{np}(t_{pm_p}) \end{bmatrix}$$

Penalized Smooth MFPCA

- Standard FPCA loadings may be noisy; smoothness penalties (via block-diagonal Ω_j) improve structure and interpretability.
- Let $\boldsymbol{X} \in \mathbb{R}^{n \times M}$ denote multivariate functional data, where $M = \sum_{j=1}^{p} m_j$.
  Its best rank-one approximation is $\boldsymbol{X} \approx \boldsymbol{u}\boldsymbol{v}^\top$,
  with $\boldsymbol{u} \in \mathbb{R}^n$ (score vector) and $\boldsymbol{v} = [v_1, v_2, \ldots, v_p]^\top \in \mathbb{R}^M$ (loading vector).
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- Define the weighted block-diagonal penalty matrix as $\boldsymbol{\Omega}_{\boldsymbol{\alpha}} = \mathrm{diag}(\alpha_1\Omega_1, \alpha_2\Omega_2, \ldots, \alpha_p\Omega_p)$, where each $\Omega_j \in \mathbb{R}^{m_j \times m_j}$ is a univariate roughness penalty matrix and $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)^\top$ contains the smoothing parameters.

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- The penalized reconstruction error is

$$\min_{\boldsymbol{u},\boldsymbol{v}} \|\boldsymbol{X} - \boldsymbol{u}\boldsymbol{v}^\top\|_F^2 + \boldsymbol{v}^\top \boldsymbol{\Omega}_{\boldsymbol{\alpha}} \boldsymbol{v}.$$

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

1. Initialize $\boldsymbol{v}$.
2. Repeat until convergence:
   1. $\boldsymbol{u} \leftarrow \boldsymbol{X}\boldsymbol{v}$
   2. $\boldsymbol{v} \leftarrow \boldsymbol{S}(\alpha)\boldsymbol{X}^\top \boldsymbol{u}$
   3. $\boldsymbol{v} \leftarrow \boldsymbol{v}/\|\boldsymbol{v}\|$
3. Update $\boldsymbol{X} \leftarrow \boldsymbol{X} - \sigma \boldsymbol{u}\boldsymbol{v}^\top$ to extract the next PC.

- The smoothing operator is $\boldsymbol{S}(\boldsymbol{\alpha}) = (\boldsymbol{I} + \boldsymbol{\Omega}_{\boldsymbol{\alpha}})^{-1} \in \mathbb{R}^{M \times M}$.

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- The smoothing operator is $\boldsymbol{S}(\boldsymbol{\alpha}) = (\boldsymbol{I} + \boldsymbol{\Omega}_{\boldsymbol{\alpha}})^{-1} \in \mathbb{R}^{M \times M}$.
- The smoothing parameter $\boldsymbol{\alpha}$ is selected via generalized cross-validation (GCV), defined as

$$\mathrm{GCV}(\boldsymbol{\alpha}) = \frac{1}{M} \frac{\|(\boldsymbol{I} - \boldsymbol{S}(\boldsymbol{\alpha}))(\boldsymbol{X}^T \boldsymbol{u})\|^2}{\left(1 - \frac{1}{M}\mathrm{tr}\{\boldsymbol{S}(\boldsymbol{\alpha})\}\right)^2}. \tag{2}$$

Penalized Sparse Multivariate FPCA

- **Goal:** Extend sparse FPCA to multivariate functional data, imposing **sparsity** (select important regions) and **smoothness** (reduce noise).
- **Sparsity penalties:** Soft, hard, or SCAD thresholding Shen and Huang [2008], Zhenhua Lin and Wang [2017], Nie and Cao [2020].

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- Sparsity parameters: $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)$, where γ_j ranges from 0 (no sparsity) to m_j for each variable.

Algorithm: Regularized Power Iteration for Sparse MFPCA

1. Initialization: Compute rank-one SVD of $\boldsymbol{X}$.
2. Iterate until convergence:
   1. Update left singular vector: $\boldsymbol{u} \leftarrow \boldsymbol{X}\boldsymbol{v}$
   2. Update right singular vector: $\boldsymbol{v} \leftarrow \boldsymbol{h}_{\boldsymbol{\gamma}} \boldsymbol{X}^\top \boldsymbol{u}$
   3. Normalize right singular vector: $\boldsymbol{v} \leftarrow \dfrac{\boldsymbol{v}}{\|\boldsymbol{v}\|}$

Smooth and Sparse Multivariate FPCA

- The combined implementation of smoothness and sparsity on the loading vector $\boldsymbol{v}$ in multivariate functional data is achieved by the following algorithm:

Algorithm: Regularized Power Iteration for Smooth + Sparse MFPCA

1. Initialize unit vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ using SVD of $\boldsymbol{X}$.
2. Repeat till convergence
   1. $\boldsymbol{u} \leftarrow \boldsymbol{X}\boldsymbol{v}$
   2. $\boldsymbol{v} \leftarrow \boldsymbol{S}(\alpha)\boldsymbol{h}(\gamma_{\boldsymbol{v}})\boldsymbol{X}^\top \boldsymbol{u}$
   3. $\boldsymbol{v} \leftarrow \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|}$
3. Update $\boldsymbol{X} = \boldsymbol{X} - \sigma \boldsymbol{u}\boldsymbol{v}^\top$ and proceed to find the next principal component.

- Algorithm CV Tuning for Sparsity and equation (2) are used to tune the sparsity level via K-fold CV and the smoothing parameter via GCV, respectively.

Simulation: Estimation Performance

Scenarios tested:

1. Unpenalized Multivariate SVD (baseline)
2. Smoothed Multivariate SVD (smoothness penalty)
3. Sparse Multivariate SVD (sparsity penalty)
4. Sparse + Smoothed Multivariate SVD (combined regularization)

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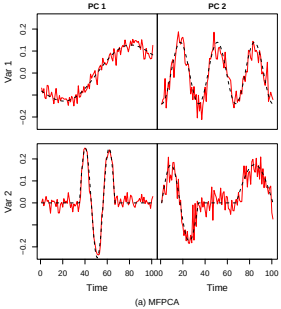

(a) MFPCA

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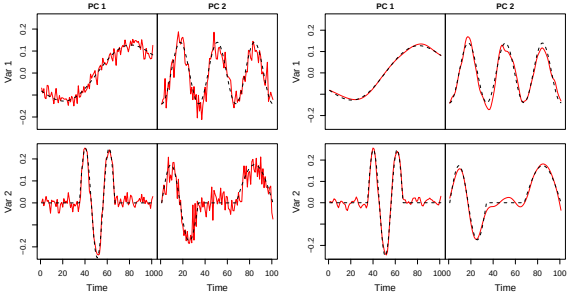

(a) MFPCA (b) Smoothed MFPCA

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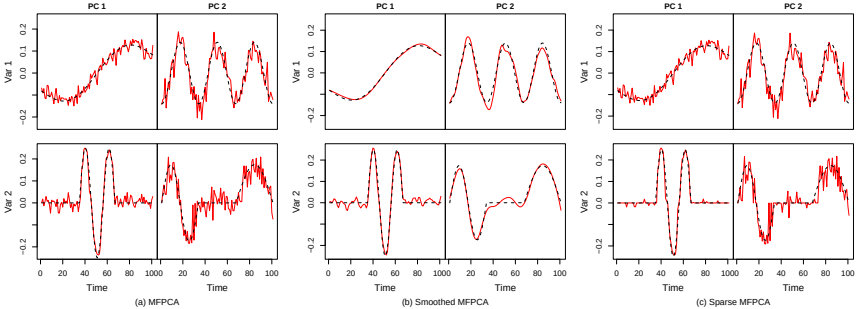

(a) MFPCA (b) Smoothed MFPCA (c) Sparse MFPCA

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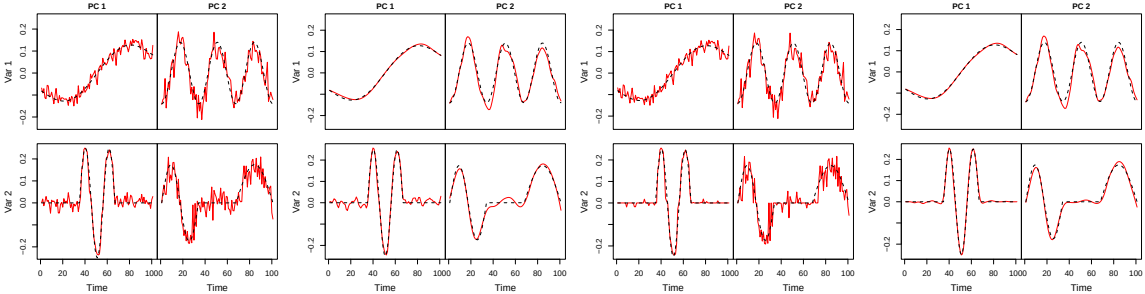
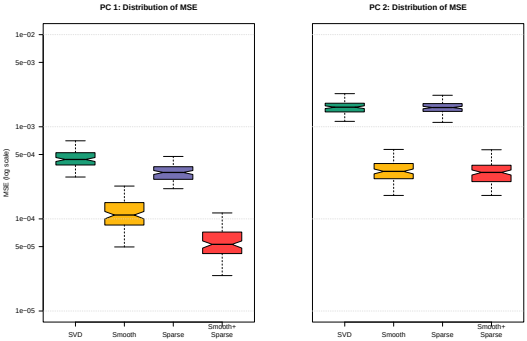


(a) MFPCA (b) Smoothed MFPCA (c) Sparse MFPCA (d) Smoothed and Sparse MFPCA

Simulation: Estimation Performance

- **Performance across four methods (SVD, Smooth, Sparse, Smooth+Sparse):**
  - Smoothness and/or sparsity **reduce MSE** compared to unregularized SVD.
  - **Smooth+Sparse** yields lowest error and most stable estimates.
  - Smooth estimator performs consistently well; sparsity alone less effective (esp. for PC2).
  - Joint regularization achieves best **bias–variance tradeoff**.



PC1: Quartiles and Mean log10(MSE)

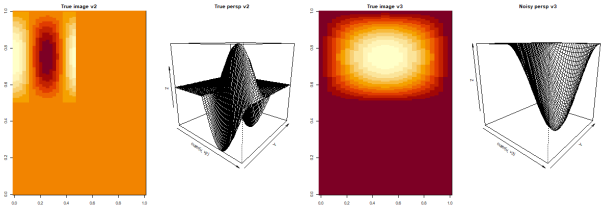
Method	Q1	Median	Mean	Q3
SVD	-3.41	-3.35	-3.34	-3.28
Smooth	-4.07	-3.96	-3.92	-3.82
Sparse	-3.57	-3.50	-3.49	-3.43
Smooth+Sparse	**-4.38**	**-4.28**	**-4.22**	**-4.14**

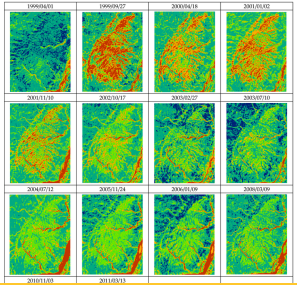
PC2: Quartiles and Mean log10(MSE)

Method	Q1	Median	Mean	Q3
SVD	-2.84	-2.79	-2.79	-2.75
Smooth	-3.56	-3.48	-3.47	-3.40
Sparse	-2.83	-2.79	-2.79	-2.75
Smooth+Sparse	**-3.59**	**-3.50**	**-3.49**	**-3.42**

Two-way Regularized MFPCA

- **Two-way functional data:** A data matrix whose row and column domains are both structured. Observations vary along these two domains, with applications in climate science, neuroscience, etc.
- **Limitation of standard FPCA [Ramsay and Silverman, 2005]:**
  - Focuses on one domain (often time).
  - Penalties applied only to loadings → ignores structure in second domain.
  - Results may be rough or overly dense along the unpenalized axis.





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  - Focuses on one domain (often time).
  - Penalties applied only to loadings $\rightarrow$ ignores structure in second domain.
  - Results may be rough or overly dense along the unpenalized axis.
- **Two-way FPCA [Jianhua Z. Huang and Buja, 2009]:**
  - Introduced **smoothness penalties** on both directions.
  - Produces coherent, interpretable *component surfaces* instead of jagged approximations.
- **Our contribution:**
  - Extend to **two-way multivariate functional data** (multiple functional variables).
  - Combine **smoothness + sparsity penalties** in both directions.
  - Result: Low-rank, interpretable, noise-robust PCs for high-dimensional applications.

Two-way Smoothed MFPCA: Setup & Penalty

- Two-way multivariate functional data: $\boldsymbol{X} \in \mathbb{R}^{n \times M}, \quad M = \sum_{j=1}^{p} m_j$.
- Roughness matrices: $\boldsymbol{\Omega_u} \in \mathbb{R}^{n \times n}$ and multivariate loading penalty $\boldsymbol{\Omega_{\alpha_v}} = \mathrm{diag}(\alpha_1 \Omega_1, \ldots, \alpha_p \Omega_p) \in \mathbb{R}^{M \times M}$, both symmetric and non-negative definite.
- Smoothers: $\boldsymbol{S_u}(\alpha_u) = (I + \alpha_u \boldsymbol{\Omega_u})^{-1}, \qquad \boldsymbol{S_v}(\boldsymbol{\alpha_v}) = (I + \boldsymbol{\Omega_{\alpha_v}})^{-1}$.

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- Penalized rank-one reconstruction:

$$\min_{\boldsymbol{u},\boldsymbol{v}} \|\boldsymbol{X} - \boldsymbol{u}\boldsymbol{v}^\top\|_F^2 + \mathcal{P}(\boldsymbol{u}, \boldsymbol{v})$$

- Penalty [Jianhua Z. Huang and Buja, 2009]:
$\mathcal{P}(\boldsymbol{u}, \boldsymbol{v}; \alpha_u, \boldsymbol{\alpha_v}) = \boldsymbol{u}^\top(\alpha_u \boldsymbol{\Omega_u})\boldsymbol{u}\, \|\boldsymbol{v}\|^2 + \|\boldsymbol{u}\|^2\, \boldsymbol{v}^\top \boldsymbol{\Omega_{\alpha_v}} \boldsymbol{v} + \boldsymbol{u}^\top(\alpha_u \boldsymbol{\Omega_u})\boldsymbol{u}\; \boldsymbol{v}^\top \boldsymbol{\Omega_{\alpha_v}} \boldsymbol{v}$.

Two-way Smooth + Sparse MFPCA

- **Goal:** Extract components that are **low-rank, smooth, and sparse**.
  - Smoothness → coherent variation across subjects & functions.
  - Sparsity → highlights key observations & time regions.
- **Novelty:** First framework to combine **both** in two-way functional data.

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- **Goal:** Extract components that are **low-rank, smooth, and sparse**.
  - Smoothness $\rightarrow$ coherent variation across subjects & functions.
  - Sparsity $\rightarrow$ highlights key observations & time regions.
- **Novelty:** First framework to combine **both** in two-way functional data.
- Data matrix $\boldsymbol{X}$: seek $\boldsymbol{u}, \boldsymbol{v}$ solving:

$$\min_{\boldsymbol{u},\boldsymbol{v}} \|\boldsymbol{X} - \boldsymbol{u}\boldsymbol{v}^\top\|_F^2 + \sum_{\ell}^{L} \mathcal{P}_\ell^{[\theta]}(\boldsymbol{u}, \boldsymbol{v})$$

- L is the number of penalty components, and θ is the vector of all tuning parameters.
- The composite penalty $\sum_{\ell=1}^{L} \mathcal{P}_\ell^{[\theta]}(\boldsymbol{u}, \boldsymbol{v})$ lets us mix regularizers, e.g., smoothness with $\theta = (\alpha_{\boldsymbol{u}}, \boldsymbol{\alpha}_{\boldsymbol{v}})$ and sparsity with $\theta = (\gamma_{\boldsymbol{u}}, \boldsymbol{\gamma}_{\boldsymbol{v}})$ (controlling sparsity), and can include other structures as needed.

Sequential Power Algorithm

Algorithm: Two-way Smooth + Sparse MFPCA (Sequential Power)

1. Initialization: Rank-one SVD of $\boldsymbol{X}$.
2. Repeat until convergence:
   1. $\boldsymbol{u} \leftarrow \boldsymbol{S}_{\boldsymbol{u}}^{[\alpha_{\boldsymbol{u}}]}\, \boldsymbol{h}_{\boldsymbol{u}}^{[\gamma_{\boldsymbol{u}}]}(\boldsymbol{X}\,\boldsymbol{v})$
   2. $\boldsymbol{v} \leftarrow \boldsymbol{S}_{\boldsymbol{v}}^{[\alpha_{\boldsymbol{v}}]}\, \boldsymbol{h}_{\boldsymbol{v}}^{[\gamma_{\boldsymbol{v}}]}(\boldsymbol{X}^\top \boldsymbol{u})$
   3. $\boldsymbol{v} \leftarrow \boldsymbol{v}/\|\boldsymbol{v}\|$
3. $\boldsymbol{X} \leftarrow \boldsymbol{X} - \sigma\, \boldsymbol{u}\, \boldsymbol{v}^\top$ to extract the next component.

- Smoothness parameters are selected with conditional GCV, while sparsity parameters are chosen via cross-validation (CV).

Selection of Regularization Parameters

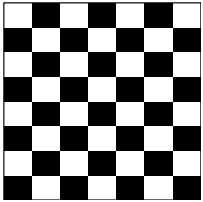
- Four sets of tuning parameters:
  - $\alpha_{\boldsymbol{u}}$: smoothness of $\boldsymbol{u}$, $\gamma_{\boldsymbol{u}}$: sparsity of $\boldsymbol{u}$
  - $\boldsymbol{\alpha_v}$: smoothness of $\boldsymbol{v}$, $\boldsymbol{\gamma_v}$: sparsity of $\boldsymbol{v}$
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.

Selection of Regularization Parameters

- Four sets of tuning parameters:
  - $\alpha_{\boldsymbol{u}}$: smoothness of $\boldsymbol{u}$, $\gamma_{\boldsymbol{u}}$: sparsity of $\boldsymbol{u}$
  - $\boldsymbol{\alpha_v}$: smoothness of $\boldsymbol{v}$, $\boldsymbol{\gamma_v}$: sparsity of $\boldsymbol{v}$
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.
- **Strategy: Conditional tuning**
  1. Initialize all penalties at 0.
  2. Tune $\gamma_{\boldsymbol{u}}$ via K-fold CV.
  3. Sequentially tune $\gamma_{v,i}$ using Algorithm:Two-way Smooth + Sparse MFPCA.
  4. With sparsity fixed, tune $\alpha_{\boldsymbol{u}}$ by GCV.
  5. Tune $\alpha_{v,i}$ using two-way GCV.
  6. Iterate steps 2–5 until stable.
- This alternating scheme **isolates sparsity vs. smoothness** while ensuring accuracy + interpretability.

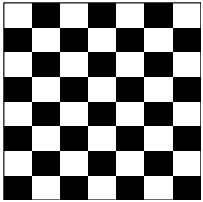
Chessboard

True Chessboard

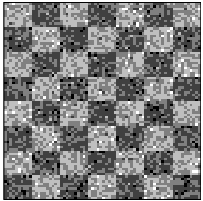


Chessboard

True Chessboard

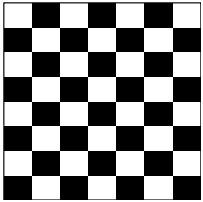


Noisy Chessboard

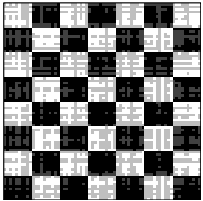


Chessboard

True Chessboard

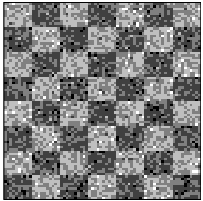


PCA with SVD

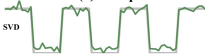


First PC Scores (u) Comparison

Noisy Chessboard

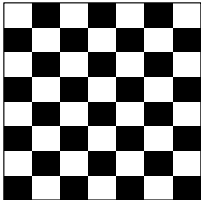


First PC (v) Comparison

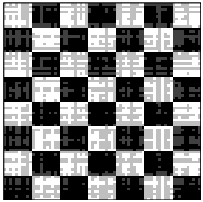


Chessboard

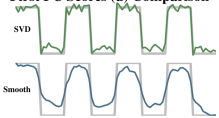
True Chessboard



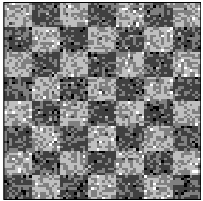
PCA with SVD



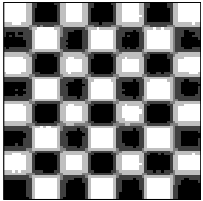
First PC Scores (u) Comparison



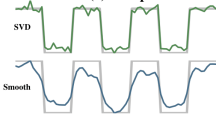
Noisy Chessboard



PCA with Smoothness Penalty

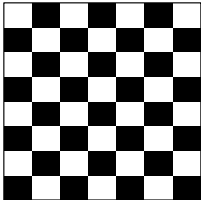


First PC (v) Comparison

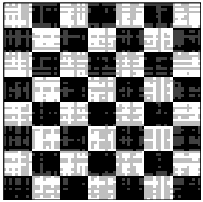


Chessboard

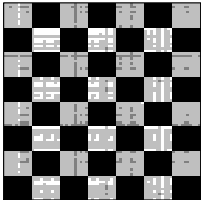
True Chessboard



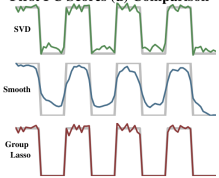
PCA with SVD



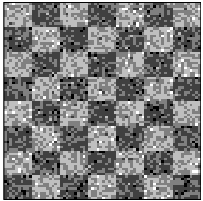
PCA with Group Lasso Penalty



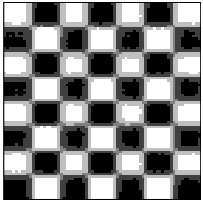
First PC Scores (u) Comparison



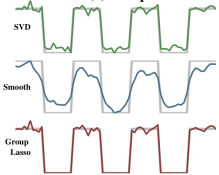
Noisy Chessboard



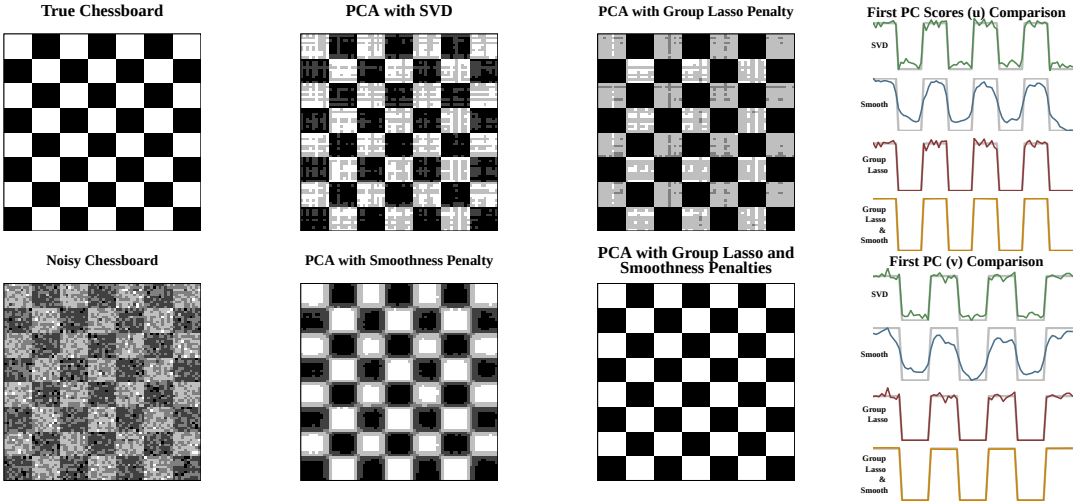
PCA with Smoothness Penalty



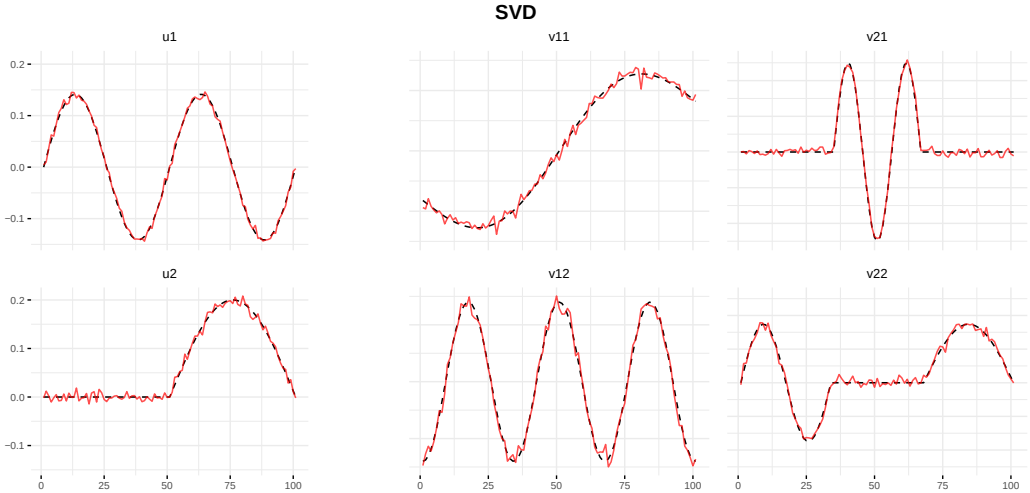
First PC (v) Comparison



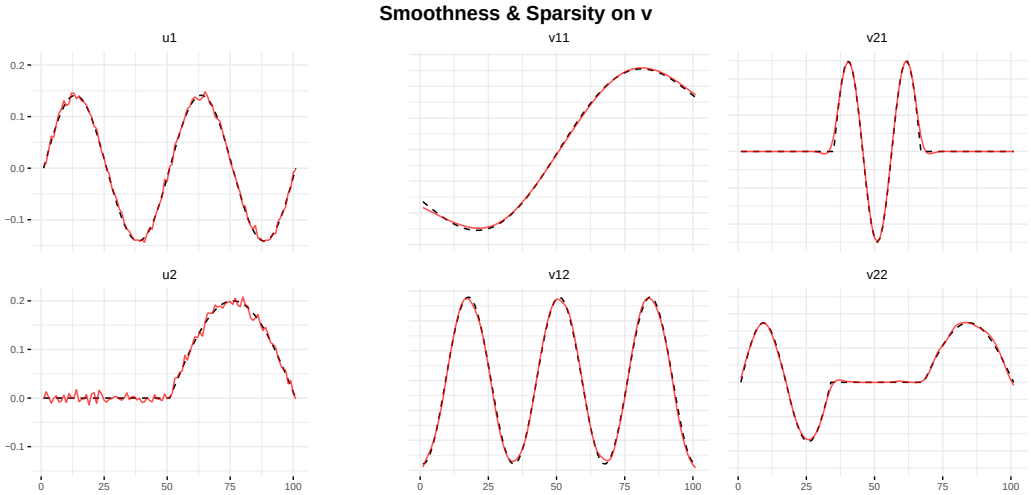
Chessboard


True Chessboard
PCA with SVD
PCA with Group Lasso Penalty
First PC Scores (u) Comparison
SVD
Smooth
Group Lasso
Group Lasso & Smooth
Noisy Chessboard
PCA with Smoothness Penalty
PCA with Group Lasso and Smoothness Penalties
First PC (v) Comparison
SVD
Smooth
Group Lasso
Group Lasso & Smooth

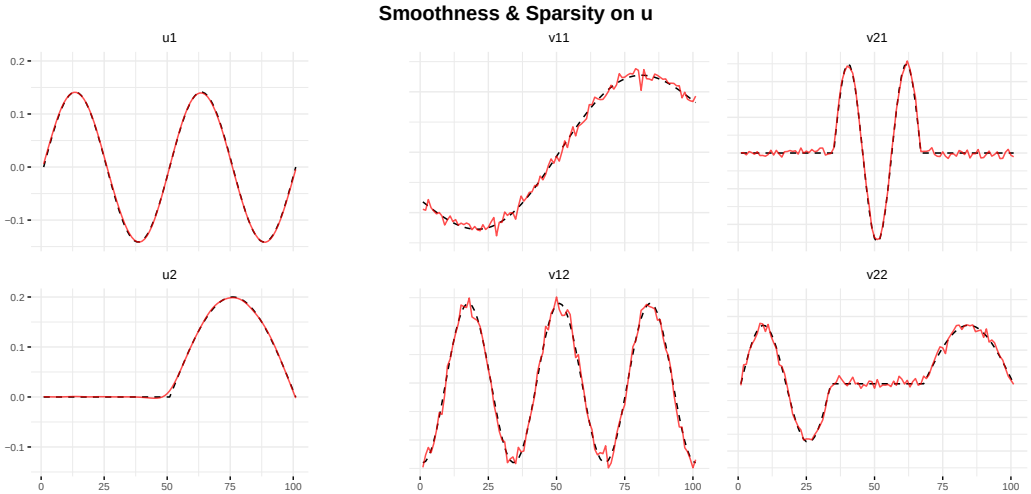
Simulation Results (SVD)



Simulation Results (Smoothness and Sparsity on v)

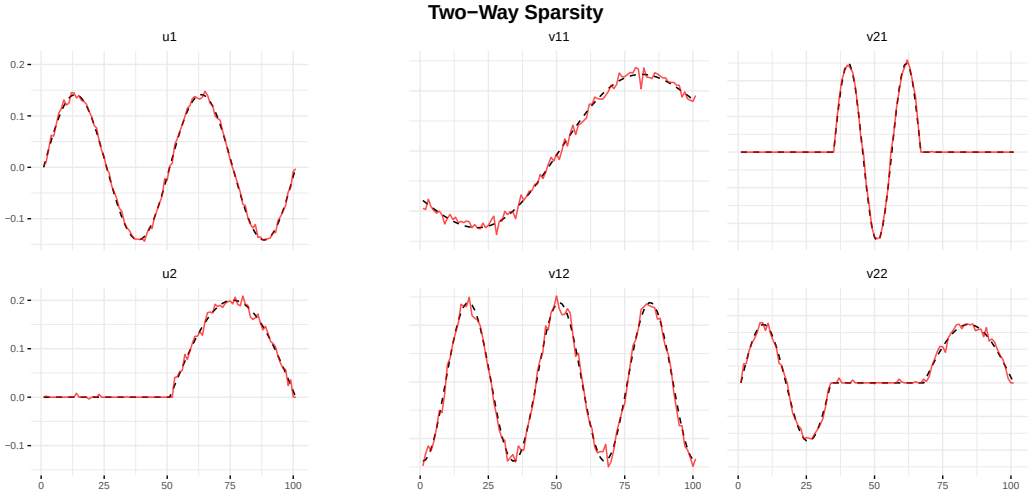


Simulation Results (Smoothness and Sparsity on u)

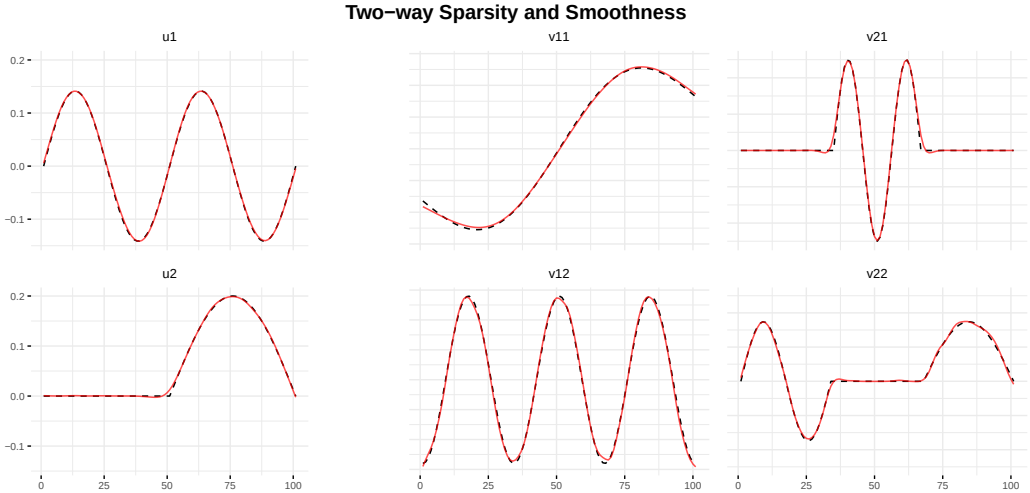


Simulation Results (Two-Way Smoothness)

Simulation Results (Two-Way Sparsity)



Simulation Results (Two-way Sparsity and Smoothness)



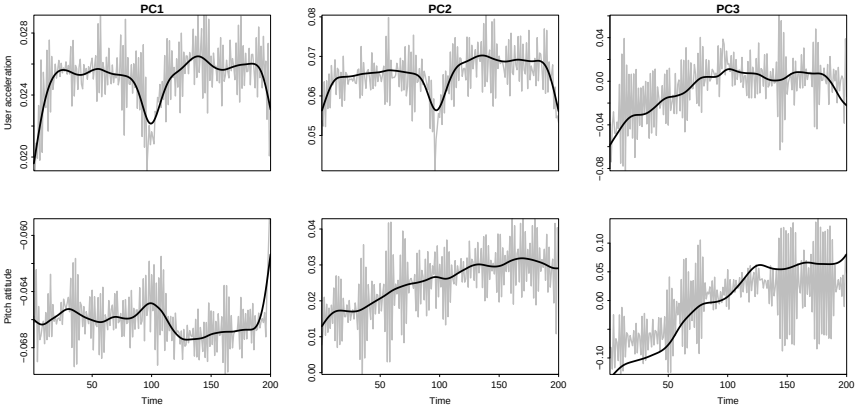
Application: Motion Sense Data

- **Dataset:** Acceleration and pitch from 24 people, 4 activities (jogging, walking, sitting, standing), about 2–3 min each.
- **Goal:** Compare SVD vs **two-way sparse + smooth ReMFPCA** on these multivariate functional signals.
- **Rescaling [Happ and Greven, 2018]:** balance variables so each contributes equally.

$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^{m} \widehat{\mathrm{Var}}(X_j(t_i)) \right)^{-1}, \qquad \tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i).$$

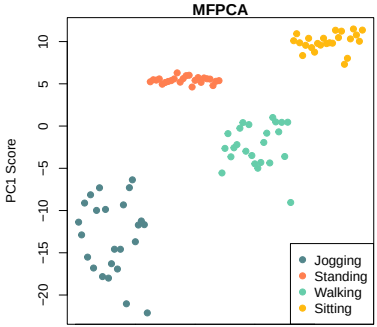
- **Penalties used:** smoothness + sparsity on loadings $\boldsymbol{v}$; sparsity on scores $\boldsymbol{u}$.

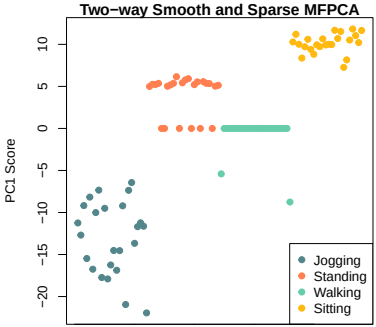
Results: Functional PCs (SVD vs ReMFPCA)



- **SVD (gray):** noisy, high-frequency wiggles.
- **ReMFPCA (black):** smoother, more interpretable PCs capturing dominant structure.

Results: PC Scores and Interpretation





- Sparsity on scores: **PC1 scores for walking about 0** (partially standing too).
- Interpretation: walking contributes little to PC1; removing it **improves interpretability** without hurting fit.
- Takeaway: **Two-way smooth + sparse ReMFPCA** yields cleaner PCs and activity-informative scores.

Conclusion & Future Work

- **Unified FPCA Framework**
  - Combines **smoothness** (denoise, interpretability) + **sparsity** (variable selection).
  - Extends from **univariate** → **multivariate** → **two-way functional data**.
- **Methodology**
  - Penalized SVD with roughness + ℓ_1 penalties.
  - Two-way regularization: smoothness & sparsity on both **scores ($\boldsymbol{u}$)** and **loadings ($\boldsymbol{v}$)**.
  - Efficient parameter tuning: **conditional GCV** & **K-fold CV** (with 1-SE rule).
- **Results**
  - Simulations & applications (mortality, call-center, image data).
  - Outperforms one-way or single-penalty methods.
  - Produces **low-rank, denoised, interpretable components**.

Accessible Implementation: R Package & Future Work

- Implemented in **R package `ReMPCA` (GitHub)**
  - Univariate & multivariate FPCA with penalties.
  - Two-way MFPCA for matrix-valued functions.
  - Automated tuning (CV, GCV, 1-SE rule).
  - Diagnostic tools: variance explained, visualization, heatmaps.
  - Early support for **hybrid data (scalar + functional + image)**.
- **Hybrid Data Extensions**
  - Image–Functional Hybrid PCA $\rightarrow$ simultaneous dimension reduction.
  - Scalar–Functional Integration $\rightarrow$ joint low-dim space.
  - Nonlinear Extensions $\rightarrow$ kernel FPCA, neural nets.
- **Applications:**
  - Neuroimaging
  - Personalized medicine
  - Environmental monitoring

  Takeaway: Smooth + sparse + two-way FPCA offers a **theoretical foundation, practical algorithms, and open software** to enable next-generation functional data analysis.

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Thank you!

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